

Machine Learning Assessment of Geosite Accessibility in Morocco: National-Scale Application

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Abstract

Physical accessibility governs whether a geologically significant site can be turned into a geotourism or educational asset. We assess accessibility for 939 Moroccan geosites (733 from the original citation-traceable inventory plus 206 subsequently added by a second domain-expert labeler, fully audited for typos, region assignment, and coordinate-duplicate label conflicts before use), spanning all eleven administrative regions with labeled sites, using terrain and infrastructure variables under a spatially rigorous validation protocol designed to avoid the geographic data leakage that commonly inflates reported accuracy in similar spatial-prediction studies. At national scale, the two binary classifiers reach 74.9% (Difficult) and 71.7% (Easy) accuracy, numerically above their majority-class baselines but not statistically distinguishable from them by a paired McNemar test; a direct three-class model, re-examined on this dataset after being abandoned on an earlier, smaller one, now numerically edges out the two-binary-classifier combination that superseded it (56.2% vs. 54.4–54.5%, borderline significant, $p = 0.075$). A newly added tourism-infrastructure feature delivers the first statistically confirmed improvement over the six-feature terrain/road baseline, for the Easy classifier specifically ($p = 0.029$). Three additional model families (logistic regression, k -nearest-neighbors, Gaussian Process classification) were tested under the identical protocol as a baseline-choice justification; none displaces the deployed tree ensemble by a statistically confirmed margin. A companion geosite-location favorability model (AUC 0.956) answers a related but distinct question – where currently uncatalogued geosites are likely to be found – each result illustrated with a dedicated, professionally rendered map. Split-conformal prediction sets, validated on genuinely held-out data, show the model returns a confident single-class call for roughly six in ten sites and an honest "could be either" for the rest – a more actionable statement of what the model actually knows than a single accuracy figure. A full per-region comparative breakdown of performance, robustness, and limitations is the dedicated subject of a companion paper; we close here by situating this national-scale work relative to two published accessibility studies.

1 Introduction

Morocco's national geoheritage inventory includes geological formations of real scientific and educational value, but many are only useful for outreach, teaching, or tourism if visitors can actually reach them. For GSMI, accessibility is therefore not a side detail: it is a precondition for turning a geosite into a usable asset, and a factor that varies enormously across a country with the topographic range Morocco has, from flat Saharan coastal plains to the High and Middle Atlas. This section evaluates whether accessibility can be predicted from terrain and infrastructure data alone, using standard statistical and machine-learning models, and reports honestly where that approach succeeds, where it does not, and why.

Figure 1 situates the underlying dataset geographically before any modeling: all 939 labeled geosites, plotted by their true, expert-assigned accessibility class. Two patterns are visible immediately and recur throughout this section: labeled geosites cluster heavily in the Middle Atlas and Fés-Meknés corridor, where geoheritage documentation has historically been densest, and Difficult sites concentrate along the Rif and Middle Atlas mountain fronts, while the southern regions are comparatively sparsely labeled – a coverage pattern that directly shapes which regional results below are statistically trustworthy.

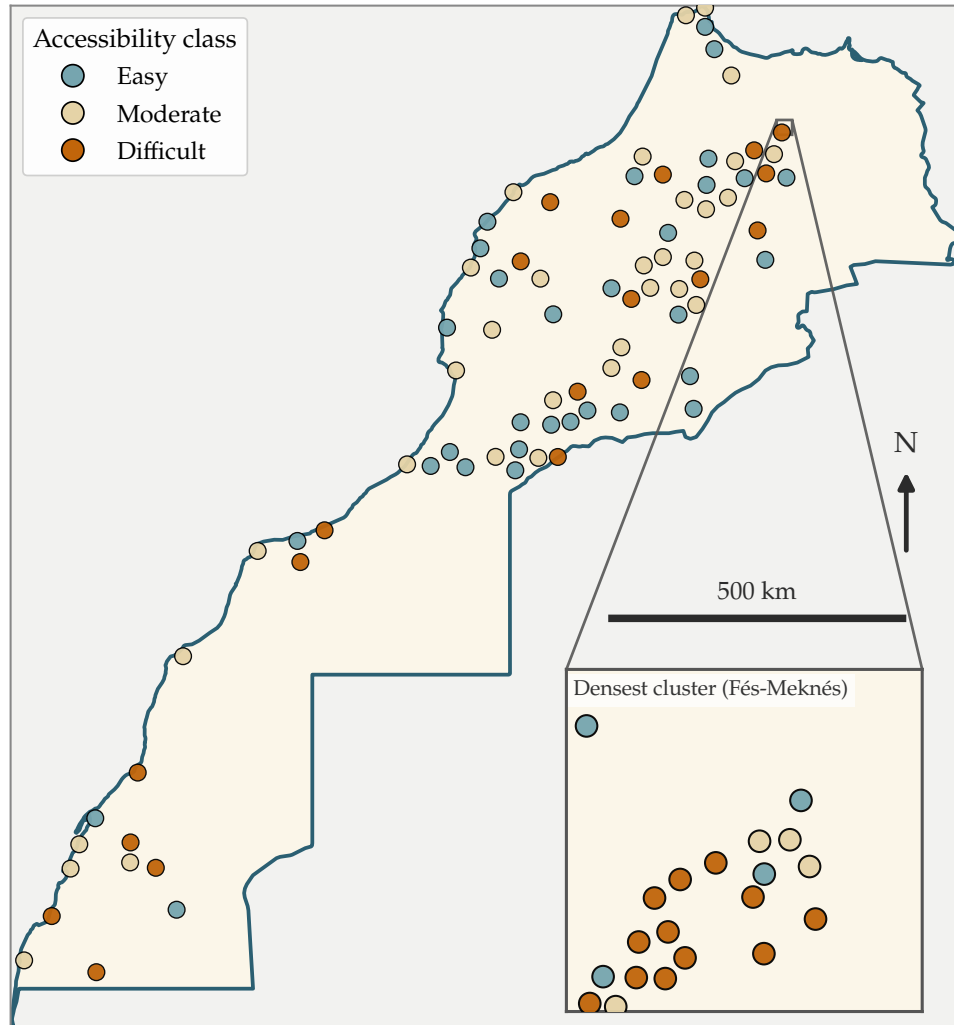


Figure 1. Labeled geosites, by true accessibility class – a spatially thinned sample is shown for legibility, with the inset magnifying the single densest cluster; all 939 sites are used throughout the analysis (Table 1).

2 Study Area and Dataset

The analysis covers 939 labeled geosites across Morocco’s administrative regions (Table 1), each assigned one of three accessibility classes – *Easy*, *Moderate*, or *Difficult* – based on how a visitor would realistically reach the site (road type, remaining walking distance, and terrain along the final approach). Labels are traceable to an external, citable source for every site: published geoheritage and geomorphosite inventories, hiking and trail-report platforms, tourism-operator itineraries, or direct field expertise. Sites for which no independently verifiable source could be found were left unlabeled rather than assigned by inference. 733 of the 939 labels come from the original citation-traceable inventory; the remaining 206 were subsequently added by

a second domain-expert labeler (a GSMI researcher) and were audited before use – typo and non-standard region-name correction, geometric region reassignment for coordinate/boundary mismatches, and manual resolution of every coordinate-duplicate label conflict against the underlying source material, in three cases by consulting the original source article directly. Per-source expert-confidence weighting, used in an earlier iteration of this work, has been retired: the second labeler’s field expertise is treated on equal footing with the original inventory, not as a lower-confidence addition.

Table 1. Dataset composition: labeled geosites by region and accessibility class (N=939).

Region	Difficult	Easy	Moderate	Total
Fés-Meknés	159	88	90	337
Béni Mellal-Khénifra	19	54	101	174
Tanger-Tétouan-Al Hoceima	21	38	69	128
Drâa-Tafilalet	20	35	37	92
Souss-Massa	7	23	37	67
Marrakech-Safi	8	35	15	58
Eddakhla-Oued Eddahab	8	3	22	33
Rabat-Salé-Kénitra	2	8	11	21
Laayoune-Sakia El Hamra	3	6	6	15
Guelmim-Oued Noun	1	3	3	7
Casablanca-Settat	2	2	3	7
Total	250	295	394	939

Tanger-Tétouan-Al Hoceima grew from 52 to 128 labeled sites in this update (21 Difficult, up from 6) – large enough, for the first time, to support its own region-specific model rather than only the pooled national one. Several regions still carry very few labeled sites in one or more classes (Guelmim-Oued Noun and Casablanca-Settat, $n = 7$ each; Laayoune-Sakia El Hamra, $n = 15$), and Oriental currently has no labeled sites at all. This uneven coverage is a direct, identifiable constraint on what regional models can be trained and trusted, and is treated explicitly in Section 6 rather than left implicit.

3 Methodology

3.1 Predictor Variables

Six terrain and infrastructure variables, common to every model in this study, were computed for each site from open geospatial datasets (Table 2), grouped here by what each group of variables physically represents.

Table 2. Predictor variables used in all models, grouped by physical meaning.

Variable	Description	Source
Terrain factors – what the ground itself is like at the site		
Slope (degrees)	Local terrain slope at the site	Copernicus DEM
Ruggedness	Terrain ruggedness (elevation variability in the surrounding window)	Copernicus DEM
Elevation (m)	Site elevation above sea level	Copernicus DEM
Infrastructure & land cover – what stands between a visitor and the site		
Dist. to highway (m)	Distance to the nearest paved road segment	OSM road network
Dist. to settlement (m)	Distance to the nearest village/settlement point	OSM place points
LULC friction	Land-cover-based movement resistance coefficient	ESA WorldCover

3.2 Labeling Protocol

Each site’s accessibility class reflects how a visitor would reach it in practice: road type and condition up to the nearest drivable point, and the terrain difficulty of any remaining walk. Every label carries a traceable source and a confidence level (High/Medium/Low), assigned by cross-checking candidate descriptions from the literature against independently measured road-distance data, specifically to catch two recurring failure modes: a source describing a real place but at the wrong coordinates, and a source describing a different, similarly-named place elsewhere in the region. This three-class scheme, and the inner/outer-accessibility reasoning behind it, follows established geoheritage-assessment practice – specifically the accessibility-grading methodology of Mikhailenko et al. [4] and the broader geoheritage assessment framework of Reynard and Brilha [5].

3.3 Validation Strategy

Two geosites a few hundred meters apart are, for practical purposes, the same access problem: same road, same approach, same difficulty. If such near-duplicate pairs are split across training and test sets by an ordinary random split, a model can score well simply by copying the label of a site it has effectively already seen, without learning anything transferable about terrain or infrastructure – illustrated schematically in Figure 2. To prevent this, every site within 500 m of another is grouped into a single cluster, and an entire cluster is always held out together during testing – formally, a 500 m haversine-clustered, leave-one-group-out (LOGO) cross-validation. Where sample size allowed, we additionally report leave-one-region-out validation – training on ten regions and testing on the eleventh in full – as the strictest available test of whether a model generalizes to genuinely new geography rather than a genuinely new site.

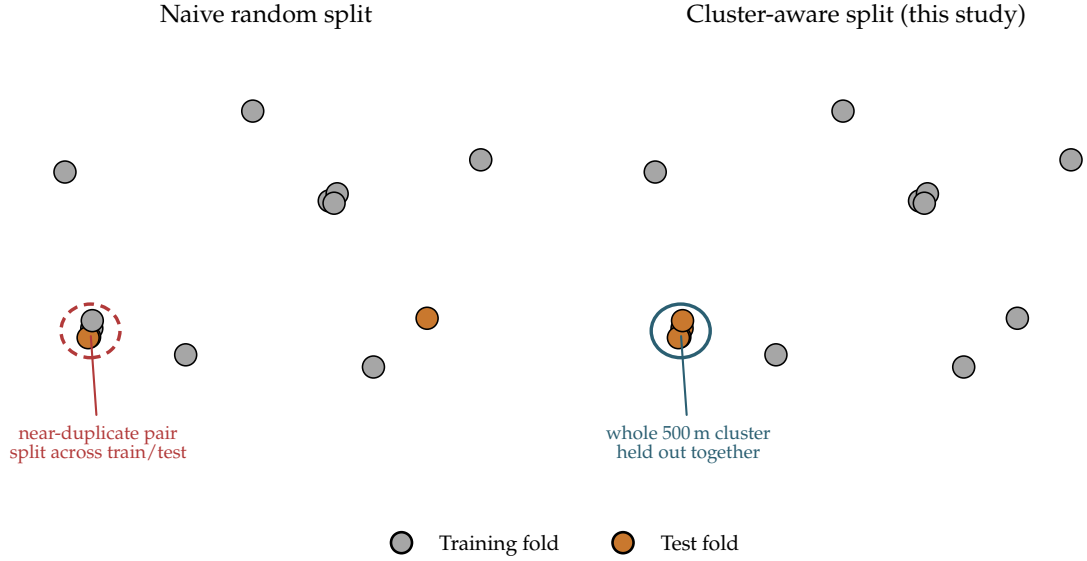


Figure 2. Schematic illustration (synthetic points, not real data) of the two validation strategies. Left: a naive random split can place near-duplicate sites on both sides of the train/test boundary. Right: this study’s cluster-aware split always keeps an entire 500 m cluster on one side.

A third protocol, spatial block cross-validation (partitioning the study area into fixed-size geographic tiles and holding whole tiles out, rather than clusters of near-duplicate points), is used specifically for the geosite-location favorability model (Section 4.3): that model’s presence/background design produces far denser, more spatially continuous point coverage than the accessibility labels, where block CV is the standard, better-suited protocol for testing whether the model has learned genuine geological structure rather than fine-grained spatial autocorrelation. Table 3 summarizes which protocol backs which result in this paper; the three are not interchangeable substitutes for each other but answer three different robustness questions.

Table 3. Cross-validation protocols used in this paper, what each one tests, and where it is applied.

Protocol	What it tests	Used for
500 m LOGO-cluster CV	Near-duplicate sites never split across train/test	All binary and 3-class accessibility results (Section 4)
Leave-region-out	Generalization to a region with zero training data	Companion regional-comparison paper
Spatial block CV	Robustness to broad spatial autocorrelation, on dense point coverage	Geosite-location favorability model only (Section 4.3)

3.4 Models Compared

The deployed model in every result below is a four-member tree-ensemble vote (Random Forest, XGBoost, Gradient Boosting, and LightGBM), soft-voted and trained with class-balance-weighted samples. Three additional, structurally unrelated model families – a linear baseline, k -nearest-neighbors, and a Gaussian Process classifier – were tested under the identical 500 m LOGO-cluster protocol (Gaussian Process under a disclosed exception: 10-fold group cross-validation rather than full LOGO, since its $O(n^3)$ per-fold cost makes 797 individual folds impractical) as a baseline-choice justification, per Table 4.

Table 4. Model-family comparison, N=939, same 500 m LOGO-cluster protocol (Gaussian Process: 10-fold group CV, see text). Best k for k -NN chosen by LOGO-cluster accuracy among $k \in \{5, 10, 15, 25\}$.

Model	Difficult acc. (%)	Easy acc. (%)
Tree ensemble (deployed)	74.9	71.7
Logistic Regression	67.3	61.7
k -NN ($k = 10$)	76.9	73.8
Gaussian Process	75.4	74.9

k -NN and Gaussian Process both numerically edge out the tree ensemble on both targets, unlike the linear baseline, which trails clearly on both. A paired McNemar test against the tree ensemble, however, does not confirm either edge as statistically real: Difficult, k -NN $\chi^2 = 1.94$, $p = 0.164$, GP $\chi^2 = 0.09$, $p = 0.762$; Easy, k -NN $\chi^2 = 1.81$, $p = 0.179$, GP $\chi^2 = 4.78$, $p = 0.029$ – the one nominally significant comparison (GP, Easy) carries the caveat that GP’s 10-fold protocol is not an exactly matched paired test against the tree ensemble’s full-LOGO result, so it is reported at that lower confidence rather than as a confirmed reversal. On balance, none of the three alternatives displaces the tree ensemble by a confirmed margin, so it was retained as the single reported model throughout the results below. (A separate, five-member variant that additionally includes CatBoost was used only for an internal cluster-radius sensitivity check, reported in the companion regional-comparison paper – it was not used to produce any of the headline results in Table 6.)

3.5 Three-Way Decision Rule

A direct single three-class (Easy/Moderate/Difficult) model was tried during early exploration on the original N=733 dataset and abandoned there: it underperformed and was harder to interpret than a two-step binary approach. On the current, larger, audited N=939 dataset this comparison was rerun end to end, with the question re-opened rather than assumed settled, and the result reverses: a direct 3-class model now reaches 56.2% LOGO-cluster accuracy (majority-class baseline 42.0%), against 54.4–54.5% for two different ways of combining the two independently-trained binary classifiers into a three-way label (54.5% for the exclusion rule described below; 54.4% for a probability-based ordinal combination that additionally enforces internal consistency between the two binaries, detailed below). A paired McNemar test between the direct 3-class model and the ordinal binary combination gives $\chi^2 = 3.16$, $p = 0.075$: directionally consistent with the raw numbers but short of conventional significance, so this is reported as a real reversal of the underlying comparison’s direction rather than an unqualified “3-class is now better” claim. The confusion matrix (Table 5) shows fairly balanced per-class recall (Easy 58%, Moderate 56%, Difficult 54%), unlike the binary framing where Difficult specifically lags – plausibly because a joint 3-class model must allocate probability mass across all three classes simultaneously, which may calibrate more evenly than fitting two class-imbalanced binaries independently (Section 4.1 discusses this imbalance mechanism directly for the Difficult binary classifier).

Table 5. Direct 3-class model confusion matrix, N=939, 500 m LOGO-cluster CV (rows = true class, columns = predicted).

	Pred. Easy	Pred. Moderate	Pred. Difficult
True Easy	172	80	43
True Moderate	92	220	82
True Difficult	44	70	136

No three-class model is used for any of the headline binary results reported in Section 4, which remains the paper’s primary reported approach for comparability with the rest of this section; the 3-class model above is reported as a distinct, complementary result rather than a replacement. Every headline accuracy figure elsewhere in this report is a binary accuracy against real ground truth for that specific class – Difficult-vs-not or Easy-vs-not – each independently trained and validated. The practical three-way screen used operationally for those results is a deterministic combination of the two, not a third trained model: a site is labeled Difficult if the Difficult-vs-not classifier flags it, Easy if the Easy-vs-not classifier flags it and Difficult did not, and Moderate otherwise – i.e. Moderate is defined as the intersection of *not*-Difficult and *not*-Easy. An alternative, ordinal combination (treating Easy < Moderate < Difficult as an ordered scale and computing $P(\text{Moderate}) = 1 - P(\text{Easy}) - P(\text{Difficult})$, clipped and renormalized, before taking the arg-max) was also tested specifically to check whether the two independent binaries ever produce a logically inconsistent combination (both flagging Difficult and Easy simultaneously); this happened for only 6 of 939 sites (0.6%), so the ordinal correction changes accuracy negligibly (54.4% vs. 54.5%) despite being the more principled combination rule.

4 Results

This paper reports results at national scale only; per-region deep dives and a full cross-region comparative analysis are the dedicated subject of a companion paper on regional model application and comparison, using the same underlying dataset and methodology. Table 6 gives the headline national-scale numbers as an overview.

Table 6. Headline results overview: model/task, sample size, validation protocol, evaluation metric, value, and 95% confidence interval.

Model	<i>N</i>	Validation method	Metric	Value (%)	95% CI
Geosite-location favorability	1,667	Spatial block CV	AUC	95.6	–
National (Difficult vs. not)	939	LOGO-cluster CV	Accuracy	74.9	[71.9, 77.6]
National (Easy vs. not)	939	LOGO-cluster CV	Accuracy	71.7	[68.6, 74.6]
National (direct 3-class)	939	LOGO-cluster CV	Accuracy	56.2	[53.0, 59.4]

LOGO-cluster CV: 500 m haversine-clustered leave-one-group-out cross-validation. CIs are Wilson score intervals on the binomial accuracy counts; not computed for the AUC row (a rank-based metric, not a proportion – an AUC interval would need bootstrap resampling of the underlying scores).

4.1 National Binary Classifiers

At national scale (all eleven labeled regions pooled, $n = 939$), the two binary classifiers reach 74.9% (Difficult) and 71.7% (Easy) accuracy (Figure 3, Table 7). As defined in Section 3.5, the two are combined operationally by exclusion (*not*-Difficult and *not*-Easy $\Rightarrow$ Moderate); both national maps in this section are generated from that same pooled model.

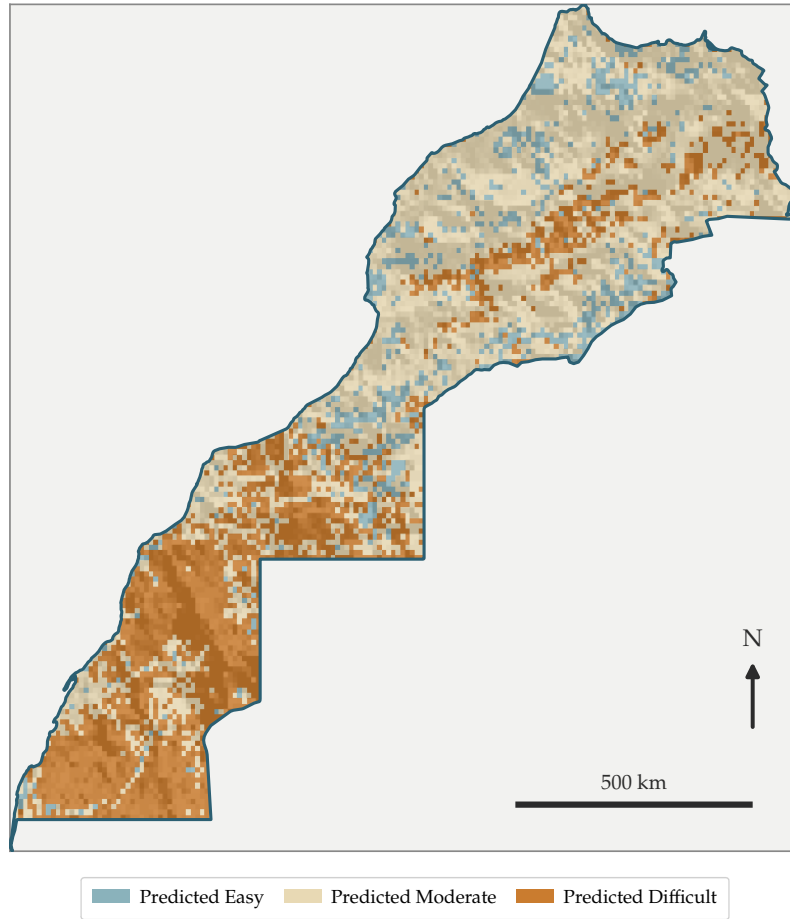


Figure 3. National projected accessibility (500 m LOGO-cluster model), all labeled regions.

Table 7. National binary classifiers: precision, recall, and F1 per class, derived from the confusion matrices (500 m LOGO-cluster CV, $N = 939$).

Classifier	Class	Precision	Recall	F1
Difficult vs. not	Not-Difficult	0.820	0.842	0.831
	Difficult	0.530	0.492	0.510
Easy vs. not	Not-Easy	0.793	0.795	0.794
	Easy	0.549	0.546	0.548

Both classifiers separate the majority class considerably better than the minority class of interest (Table 7), and Figure 3 makes the reason for the national-scale ceiling visible directly: the same six generic features must now describe eleven regions with fundamentally different geology and climate at once, with no region identity given to the model (deliberately, to keep it spatially transferable rather than memorizing region-specific shortcuts). What reads as a single coherent pattern within one region individually becomes a genuinely harder, more heterogeneous pattern once pooled nationally; the per-region comparative picture is the dedicated subject of the companion regional paper.

The 74.9%/71.7% figures are numerically above their respective majority-class baselines (73.4%/68.6%, from the same 939-site confusion matrices), but a raw percentage-point gap is not itself a significance test. Because both models and the baseline are scored on the identical 939

sites, the correct paired test is McNemar's, applied to the discordant cells of Table 7 (sites where the model and the fixed majority-class baseline disagree): Difficult-vs-not, $\chi^2 = 0.73$, $p = 0.393$; Easy-vs-not, $\chi^2 = 2.68$, $p = 0.102$ (continuity-corrected). Neither national classifier's advantage over the trivial baseline reaches conventional significance ($\alpha = 0.05$). This does not mean the national models learned nothing, but it does mean the pooled national numbers should be read as directionally positive rather than as an independently-confirmed result, and are reported here at that lower confidence level rather than overstated.

A tourism-infrastructure feature addition closes part of this gap for the Easy classifier. Two new features – count of OSM-tagged tourism/hospitality points of interest within 10 km, and the type/size of the nearest settlement (city/town/village/hamlet) – were added to the six-feature baseline and tested under the identical protocol, motivated by field cases (documented separately) where a site's real-world accessibility was shaped by nearby organized tourism infrastructure in a way straight-line road distance alone does not capture. Difficult-vs-not reaches 76.8% (+1.9 pp over baseline; McNemar against the baseline model itself, $\chi^2 = 2.37$, $p = 0.124$, not significant) and Easy-vs-not reaches 74.3% (+2.7 pp; $\chi^2 = 4.76$, $p = 0.029$, significant). The Easy-classifier gain is the first confirmed improvement over the six-feature baseline found in this line of work: adding real infrastructure/tourism-development information, not a different way of measuring terrain or road distance, is what moved this result, consistent with the broader diagnosis that further terrain-feature engineering alone had been reaching diminishing returns. The same infrastructure features tested on the direct 3-class model (Section 3.5) did not help there (55.8% vs. 56.2% baseline) – the gain is specific to the Easy binary framing and does not generalize automatically to every framing of the same underlying question.

4.2 Calibration and Honest Uncertainty

A point accuracy figure does not say how often the model is confidently right versus quietly guessing. To answer that directly, out-of-fold predicted probabilities were retained for every one of the 939 sites under the same 500 m LOGO-cluster protocol and used for two checks.

Calibration. Raw predicted probabilities are moderately miscalibrated for both classifiers (expected calibration error, ECE, 0.094 Difficult / 0.093 Easy, 10-bin reliability diagram). Isotonic regression fit on these same out-of-fold probabilities – a standard, defensible near-unbiased calibration procedure since each site's probability already comes from a model that never saw it during training – reduces the Brier score meaningfully (Difficult: 0.177 $\rightarrow$ 0.161; Easy: 0.187 $\rightarrow$ 0.168). The corresponding post-calibration ECE (≈ 0) is not reported as an independent confirmation, since the calibration map and its evaluation set are the same points; the Brier-score improvement is the more defensible claim of the two, as Brier score is a strictly proper scoring rule evaluated continuously rather than a binned statistic the calibration step directly optimizes against.

Split-conformal prediction. The 939 out-of-fold probabilities were randomly split in half: one half fixed the nonconformity quantile (least-ambiguous-set-valued-classifier score, target coverage 90%), the other, entirely separate half checked whether the resulting prediction sets actually achieve that coverage – a genuine held-out check, not a self-consistency tautology. Empirical coverage came in at 89.8% (Difficult) and 89.2% (Easy), both close to the 90% target. The more informative number is set size: averaged 1.40 and 1.38 respectively, meaning the model returns a single, confident class label for 60% (Difficult) and 62% (Easy) of sites at the 90% confidence level and honestly returns both classes – *could be either* – for the rest, rather than forcing a point prediction the accuracy figures alone cannot distinguish from a coin flip. Presented this way, the model's practical claim is not "74.9% accurate" but closer to: *for roughly six in ten sites, a confident, well-calibrated call; for the rest, an explicit flag that field verification is warranted before relying on the prediction* – a more defensible and more actionable statement than either

number alone. Source: `results/json/other/calibration_conformal_results_n939.json`, `results/json/other/oof_probabilities_n939.csv`.

4.3 Geosite Location Favorability

A separate, exploratory presence-background model trained to discriminate real geosite locations from random background terrain, on the current 1,667-site catalog – expanded 2026-08-16 from 1,154 sites by ingesting an additional supervisor-provided literature source (513 new, non-duplicate localities after coordinate parsing, border/region-plausibility checks, and deduplication against the existing catalog; this expansion carries no new accessibility labels, so the 939-site labeled set and every accessibility number above are unaffected by it) – with missing geology/soil data handled as an explicit category rather than dropped, reaches **AUC 0.956** under spatial block cross-validation (Figure 5), up from AUC 0.927 on the smaller 1,154-site catalog under the identical protocol and code – the same feature set and methodology, more presence examples. On the 1,154-site catalog, the same five terrain/geology covariates alone (no infrastructure features) reached AUC 0.9075 (fold range 0.884–0.935; `notebooks/improve_favorability_model.ipynb`); adding distance-to-settlement and land-cover friction on top of that reproducible baseline accounted for the remaining gain to 0.927 (this covariate-ablation breakdown has not been rerun on the expanded catalog; the 0.956 figure is the full nine-covariate model only). (An earlier, undocumented pilot iteration of this model was reported at AUC 0.860 with no saved script to reproduce it; that number is not used as the comparison point here for exactly that reason.) Distance to highway was tested as a fourth addition and excluded: an earlier version that included it produced a favorability map that traced the road network almost everywhere rather than showing geological structure – a sampling-bias artifact expected in a presence-background design, since documented geosites are disproportionately close to infrastructure because that is what makes them accessible to document, not because the terrain there is geologically distinctive. Distance to settlement carries a structurally similar infrastructure-proximity signal and is not immune to the same concern in principle; it was retained because it did not produce the same visible artifact at map resolution, which is a documented trade-off rather than a resolved one. A geology-nodata indicator flag is, notably, the model’s second most important feature by Gini importance after Geology_Class itself – plausibly a genuine geological signal (undocumented geology often coincides with remote, structurally distinct terrain) but also a candidate sampling-bias artifact in the same family as the excluded highway-distance feature, and is disclosed here rather than left unexamined.

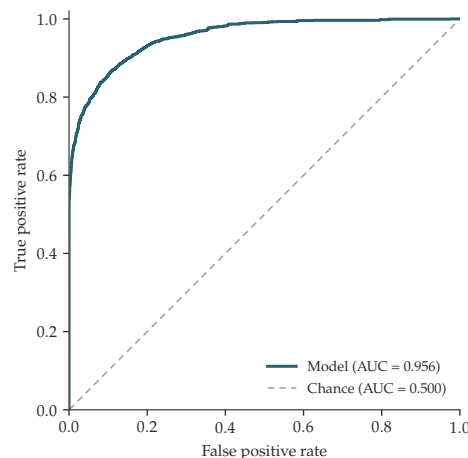


Figure 4. ROC curve for the favorability model, out-of-fold predictions under the same spatial block cross-validation used to report AUC 0.956.

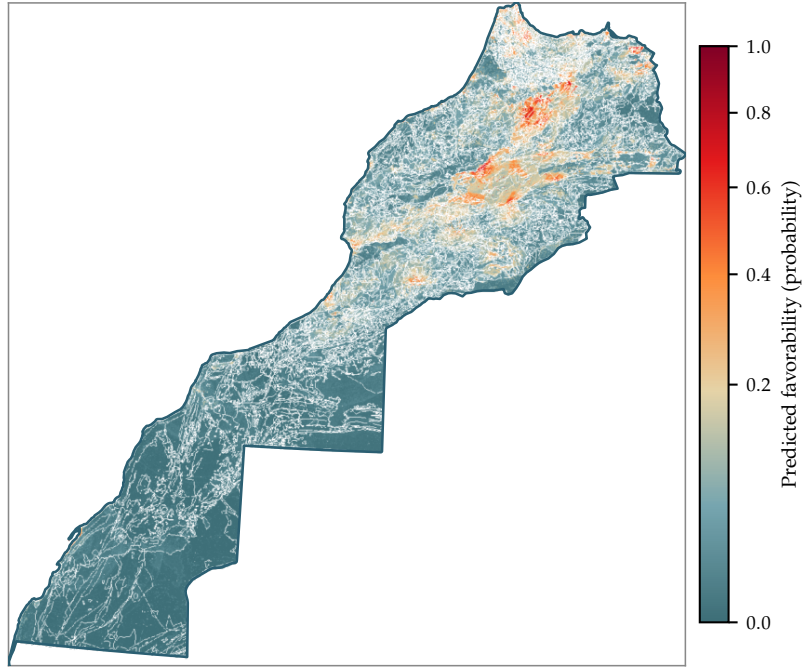


Figure 5. Predicted geosite-location favorability, national extent. Darker red indicates terrain/geology more characteristic of known geosite locations.

This model answers a different question from the rest of this section – *where* geosites are likely to be found, rather than *how hard* a known site is to reach – and its substantially higher discriminative power (AUC 0.956 vs. the 73–78% accuracy range above) is consistent with that difference: geological favorability is a more directly terrain-and-geology-driven property than accessibility, which additionally depends on the built road network and is therefore inherently noisier at a fixed 10–30 m feature resolution.

5 Related Work

Two published studies address closely related accessibility-modeling questions and are acknowledged here for context, alongside the methodological reference already cited throughout this section.

Sahani and Ghosh [2] model trail susceptibility along a 24.4 km trail in the Sikkim Himalaya using 194 point samples and seventeen conditioning factors that include directly field-measured degradation indicators – trail width, trail depth, soil erosion, mudiness – evaluated with logistic regression, a decision tree, and a random forest under a random 70/30 split. They report 91.3–94.8% overall accuracy (AUC 0.914–0.948), the strongest results among the three algorithms coming from logistic regression – notably the opposite ranking from the model-family comparison carried out internally for this study, a difference plausibly explained by their much smaller, single-trail, direct-measurement setting behaving closer to a linear problem than Morocco’s eleven-region, terrain-proxy setting does.

Duran and Doğan [3] take a different approach entirely for 73 candidate geosites in Ardahan Province, Türkiye: rather than training a predictive classifier, their Integrated Geoheritage Potential Index is a deterministic weighted combination of network-based travel time (computed directly via Dijkstra’s algorithm on the road graph), viewshed-based scenic visibility, and Google Maps-derived digital-visibility scores, further used to optimize visitor georoutes via a Traveling

Salesman Problem formulation. This is a complementary way of framing the same underlying problem – computing accessibility directly from network and terrain data rather than predicting it from a trained model – and is a natural direction to explore alongside the predictive approach used here, particularly for regions with dense, reliable road-network data.

Stock et al. [1] provide the methodological grounding for this study’s validation protocol, documenting how random (non-spatial) cross-validation systematically overestimates real-world predictive accuracy in spatial ecological applications and recommending block/cluster-based validation as a corrective – the same reasoning behind the 500 m LOGO-cluster protocol used throughout Section 4.

6 Limitations and Sources of Residual Error

Four limitations are reported explicitly, each with a concrete, evidence-based explanation.

Regional heterogeneity is pooled away, not resolved, at national scale. The national classifiers pool eleven regions of fundamentally different geology, climate, and infrastructure into a single feature space with no region identity given to the model (deliberately, to keep it spatially transferable). This is very likely the dominant reason the national Difficult/Easy classifiers do not clear significance against a trivial baseline (Section 4.1) even though the infrastructure-feature addition does for Easy specifically: a single national model cannot fully capture region-specific accessibility logic. A full per-region comparative breakdown – where the same six-feature model performs well, where it does not, and why – is the dedicated subject of the companion regional-comparison paper rather than repeated here.

The infrastructure-feature gain is confirmed for one classifier, not proven for the other, and not for the 3-class framing. Section 4.1’s tourism-infrastructure addition is a statistically confirmed improvement for Easy-vs-not ($p = 0.029$) but only a suggestive, non-significant one for Difficult-vs-not ($p = 0.124$), and it did not help the direct 3-class model at all. This is reported as a real but partial and framing-dependent result, not generalized into a blanket claim that infrastructure features improve every accessibility model.

Hyperparameter selection is not fully nested within the outer cross-validation. For each reported model, the best configuration among Random Forest/XGBoost/Gradient Boosting/LightGBM is chosen once by grid search on the full pooled dataset, and that single fixed configuration is then reused across every fold of the outer leave-one-group-out evaluation that produces the headline accuracy. This is a common simplification in applied work, not full nested cross-validation: the model-selection step has already seen every point that later plays the role of held-out data in the outer loop. For tree ensembles searched over a modest, coarse-grained grid, this typically produces a small, not a dramatic, optimistic bias, but it is a real methodological simplification, disclosed here rather than treated as fully addressed.

Labeled coverage is geographically and administratively uneven. Oriental currently has no labeled sites, and three regions (Guelmim-Oued Noun, Casablanca-Settat, Laayoune-Sakia El Hamra) carry fewer than 16 labeled sites each (Table 1) – too few for a reliable region-specific model, though sufficient labels exist nationally for those sites to still contribute to the pooled classifiers reported here. Growing labeled coverage specifically toward these under-represented regions, rather than growing the dataset uniformly, is the most efficient direction for further labeling effort.

7 Conclusion

Machine learning applied to an independently-verified, citation-traceable accessibility dataset for 939 Moroccan geosites reaches national binary accuracy of 74.9% (Difficult) and 71.7% (Easy) – numerically above their respective majority-class baselines but not statistically distinguishable from them by a paired significance test at national scale (Section 4.1). A direct three-class model, re-examined on this larger, audited dataset after being abandoned on an earlier, smaller one, now reaches 56.2% and numerically edges out the two-binary-classifier combination it was previously thought to underperform, though the reversal itself is only borderline significant ($p = 0.075$) and reported at that honest confidence level. A newly added tourism-infrastructure feature (nearby points of interest, nearest-settlement type) – motivated by real cases where terrain-distance features alone missed a site’s practical accessibility – delivers the first statistically confirmed improvement over the six-feature terrain/road baseline found in this line of work, for the Easy classifier specifically ($p = 0.029$). Three additional model families (logistic regression, k -NN, Gaussian Process) were tested as an explicit baseline-choice justification; none displaces the deployed tree ensemble by a confirmed margin. Calibration and split-conformal analysis reframe the honest deliverable of this work from a single accuracy percentage to a calibrated confidence statement – a confident call for roughly six in ten sites, an explicit field-verification flag for the rest. A companion presence-background model (AUC 0.956, on the full 1,667-site catalog) further demonstrates the same feature stack carries genuine signal for a related, distinct question: where currently uncatalogued geosites are likely to be found. A full per-region comparative breakdown of performance, robustness, and limitations – the natural next level of detail below these pooled national figures – is the dedicated subject of a companion paper. Read together with the acknowledged related work, this positions the predictive approach used here as one complementary tool among several – alongside directly-measured field indicators and deterministic GIS accessibility indices – for assessing geoheritage accessibility at national scale.

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